Type III

Technicians maintaining, servicing, repairing, or disposing of low-pressure appliances must be certified as a Type III Technician or a Universal Technician. Most low-pressure appliances are used for comfort cooling and are subject to a lower allowable leak rate.

Leak Detection

Because a low-pressure system operates below atmospheric pressure (in a vacuum), leaks in the gaskets or fittings will cause air and moisture to enter the system.

Low-pressure chillers require purge units because they operate below atmospheric pressure, Figure 16. The primary purpose of a purge unit is to remove non-condensables that leak into a low-pressure chiller system. The purge unit gets its suction from a connection on top of the condenser where most non-condensables accumulate. After separating the air from the refrigerant, the purge unit sends the refrigerant back to the evaporator. High head pressure indicates air in a low-pressure system and that the purge unit may not be operating properly. Excessive running of a purge system on a low-pressure chiller generally indicates a leaking system. High-efficiency purge units will discharge a low percentage of refrigerant with the air they remove.

Figure 16. Purge Unit.

A hydrostatic tube test kit is used to determine if a tube is leaking. When leak testing a low-pressure centrifugal chiller with nitrogen, the maximum test pressure is 10 psig. Leak testing a low-pressure refrigeration system with nitrogen in excess of 10 psig could cause the rupture disc to fail. Charged low-pressure refrigeration machines may be most efficiently leak checked by adding heat with circulated hot water or heating blankets. To reduce refrigerant loss from the purge unit on an R-123 chiller, always leak test and repair the chiller. A low-pressure refrigeration system could have a leak in the condenser or chiller barrel tubes if excessive moisture continuously collects in the purge unit.

To prevent air accumulation into an idled low-pressure refrigeration system, maintain system pressure slightly above atmospheric pressure.

Moisture most frequently enters the refrigerant system in low-pressure chillers through leaks from areas with gaskets or fittings. To check for a refrigerant leak to the water side of a chiller, place a leak detector probe in the drain valve opening of the water box with the water removed. Drain the water sides of the evaporator and condenser prior to recovering refrigerant from a chiller suspected of having tube leaks.

Low-pressure refrigeration systems with open-drive type compressors are particularly susceptible to leaks from the shaft seal. According to ASHRAE Guideline 3, the system should be checked for leaks if, during a standard vacuum test, the pressure in a system rises from 1 mm Hg to a level above 2.5 mm Hg.

Leak Repair Requirements

The EPA's leak repair requirements apply to all appliances using a regulated refrigerant. Once the threshold leak rate has been exceeded on any system using a regulated refrigerant with a charge of 50 pounds or greater, the owner must repair the appliance to bring the leak rate below the threshold, retrofit, or retire the appliance.

As of January 1, 2019, the applicable maximum leak rates are:

- 10% of the charge per year when used for comfort cooling.
- 20% of the charge per year when used for commercial refrigeration.
- 30% of the charge per year when used for industrial process refrigeration.

Chillers generally belong to the EPA leak repair category for comfort cooling. When used in multiple categories, the applicable maximum leak rate is determined by the highest percentage of use, Figure 17.

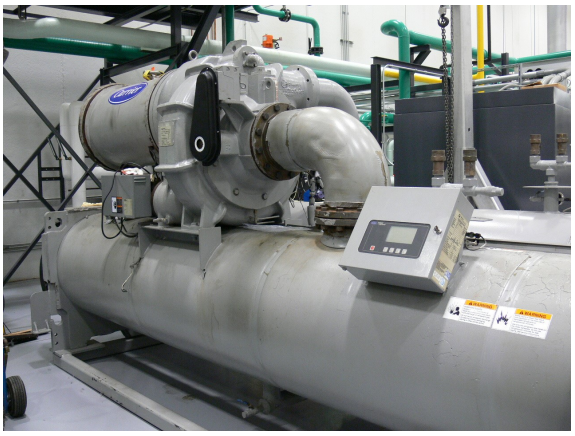

Figure 17. Low-Pressure Chiller.

Example: If a company diverts 55% of its chiller's cooling capacity to one of its manufacturing lines and 45% of its cooling capacity to air condition the office spaces, the chiller is considered in the industrial process refrigeration (IPR) leak rate category.

Use the charge stated on the equipment nameplate to determine a chiller's normal charge for leak rate calculations. The leak rate of an appliance using a regulated refrigerant must be calculated when topping off or recharging a system due to a loss of charge.

If an appliance with 50 or more pounds of a regulated refrigerant charge has exceeded the threshold leak rate, the owner or operator has 18 months to retrofit or retire the appliance if the replacement appliance will use a refrigerant exempt from the venting prohibition.

Not having a certified service technician available for service cannot be used as a reason to extend the appliance leak repair deadline.

System mothballing does not require removing the appliance and storing it in a warehouse at the facility.

If an appliance containing 50 or more pounds of a regulated refrigerant has exceeded the threshold leak rate, the owner or operator has 30 days to repair the appliance so that it leaks below the threshold and conduct the initial verification test.

If a chiller with a charge of 200 pounds of an ozone-depleting refrigerant has passed the initial verification leak test, a follow-up verification test must be conducted within 10 days.

The owner and/or operator of the equipment is responsible for and must keep records of all leak inspections, initial verification, and any completed leak repair follow-up verification tests for three (3) years.

Recovery Techniques

The high-pressure cut-out control on a recovery unit used for evacuating/recovering the refrigerant from a low-pressure chiller is typically set for 10 psig. Due to the volume of liquid refrigerant in large systems, refrigerant removal from a low-pressure system starts with liquid removal followed by recovery of the vapor refrigerant. After all the liquid has been removed, an average 350-ton, R-123 chiller, at 0 psig, would still have approximately 100 pounds of refrigerant in the vapor state.

The use of a severely oversized vacuum pump could cause trapped water to freeze when evacuating/ dehydrating a system. During evacuation/dehydration of a system containing large amounts of moisture, it may be necessary to break the vacuum using nitrogen. Increasing the pressure with nitrogen prevents the moisture from freezing and increases the speed of dehydration.

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A water-cooled recovery unit will allow for faster recovery of large quantities of refrigerant. Water is circulated through a chiller during refrigerant evacuation/ recovery to prevent freezing of water in the appliance.

The system's water pumps, recovery compressor, and recovery condenser water should all be on during vapor or liquid removal from a low-pressure refrigeration system. Raising the temperature of the system or the room in which a low-pressure chiller is located will also result in faster refrigerant recovery.

When removing refrigerant oil, it should be heated to 130°F as less refrigerant will be contained in the oil at the higher temperature. Be careful when heating the system, since the rupture disc on a recovery vessel for low-pressure refrigerants relieves at 15 psig, Figure 18.

Figure 18. Rupture Disc.

Recovery Requirements

After reaching the required recovery vacuum on a low-pressure appliance wait a few minutes to see if the system pressure rises before charging. When leaks in the appliance make evacuation/recovery to the prescribed level unattainable, do not proceed with the recovery. Recovery and recycling equipment must be labeled as certified to meet EPA's requirements.

Replacing or repairing a compressor, condenser, or evaporator is considered a major repair. When making a non-major repair, warming the liquid refrigerant in the system with hot water or warming blankets will bring the system pressure out of a vacuum.

Before disposing of a low-pressure appliance, the refrigerant must be recovered or evacuated to 25 mm of Hg absolute. Recovery records for disposed appliances with 5-50 lbs. of refrigerant must be kept for three (3) years.

Recharging Techniques

Before evacuation/recovery or dehydration, an oil sample should be taken if the unit has had a compressor burn-out. Also, when evacuating/dehydrating a system that contains a large amount of moisture, triple evacuate/dehydrate the system using dry-nitrogen to break the vacuum. Increasing the pressure with nitrogen will prevent the moisture from freezing.

After servicing and evacuating/dehydrating a chiller, refrigerant vapor is re-introduced into the chiller to a pressure which corresponds to its saturation temperature above freezing. This will prevent liquid refrigerant charged into the system from freezing water in the heat exchanger tube bundle. For example; charging liquid R-245fa into a low-pressure refrigeration system in a vacuum greater than 18 inches Hg vacuum will cause the system water to freeze. At 18 inches Hg vacuum, R-245fa has a saturation temperature well below -40°F. When reviewing the saturated conditions on a PT chart, an evacuated/dehydrated water-cooled chiller system with R-123 should have a vapor pressure above 20 inches Hg vacuum before adding liquid refrigerant. The liquid refrigerant is charged into the system using the evaporator charging valve, which is the lowest access point on a low-pressure appliance.

Note: The relationships found on pressure-temperature charts will vary slightly from one manufacturer or publisher to the next. The pressures are normally indicated in psig. To convert from absolute pressure (psia) to gauge pressure (psig) subtract 14.7 from the absolute pressure. Therefore, 14.7 psia equals 0 psig.

Safety

When the TLV-TWA (Threshold Limit Value - Time Weighted Average) is exceeded, ASHRAE Standard 15-2013 requires that each machinery room activate an alarm and a mechanical ventilation system.

ASHRAE Standard 15-2013 also requires the use of room sensors and alarms to detect refrigerant leaks in all refrigerant safety groups, Figure 19. R-123 refrigerant is classified as B1 (non-flammable, high toxicity) and refrigerant R-1233zd is classified A1 (non-flammable, low-toxicity) under Standard 34 code group. The use of room sensors and alarms to detect refrigerant leaks in all refrigerant safety groups is required not only because of refrigerant characteristics, but because refrigerants are heavier than air and can displace oxygen.

Figure 19. Equipment Room Wall Plate.

Gloves and safety goggles should be worn when working with liquid refrigerant for low-pressure systems.

The discharge from a rupture disc or pressure relief valves should be piped outdoors to prevent venting into the machine room. All pressure relief valves should be piped or installed in parallel, **never** in series.

A 15 psig rated rupture disc is located on the evaporator of a centrifugal chiller. When charging, refrigerant is introduced through the evaporator charging valve, make sure not to over-pressurize the system.

Note: Isopropyl alcohol should be used to remove ice from sight glasses or viewing glasses.

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